Smart Indoor Environment Control

Presented by Group 6
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Client, Context, and Team

Client and Context

- **Client** — Faculty and Graduate Student from BCIT Building Sciences:
  - Dr. Rodrigo Mora
  - Caio Nunes
- **Context** — Improve human and building’s interactions

Our Team

Member	Responsibilities
Kai	Database, configuration, backend coordination
Heather	UI design, integration, data processing
Shinyoung	Frontend responsiveness, backend connection
Hei	Manual device control, integration
Khai	Survey redesign, code logic refactor

Problem and Purpose

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Problem and Project Purpose

- **Problem** — Room conditions vary during the day
- **Goal** — Improve comfort and avoid energy waste
- **Project Purpose**
  - Collect feedback, sensor data
  - Display room conditions
  - Give recommendations
  - Let users manually override



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Starting Point and Scope

Starting Point and Scope

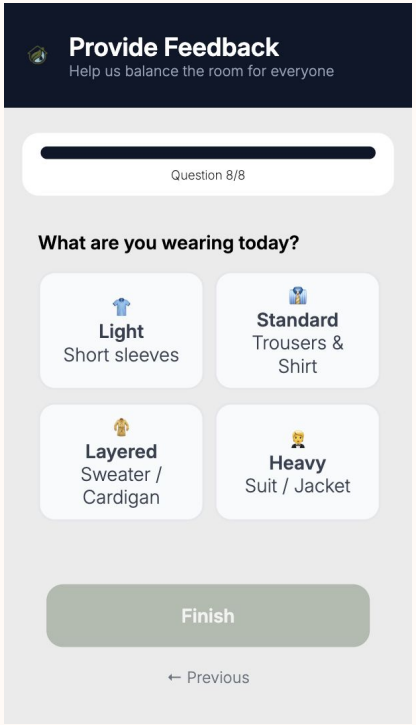
What Existed		Project Scope
Basic survey + dashboard	→	Redesign dashboard + survey
Rough database	→	Restructure database
Basic rule-based system	→	Add manual override
Mock sensor data + devices	→	Integrate devices + sensors

Our Contribution

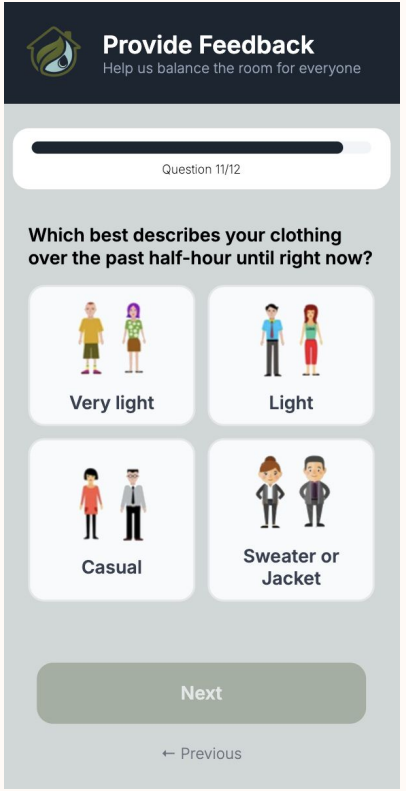
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Our Contribution

- **What was inherited**
  - Occupant can submit & save feedback
  - **Before**: Way too simple DB structure → hard to use for research
- **What our team improved + why better now**
  - Added more DB tables and attributes → easier data import for research
  - More questions (total 8 → 12) with more relevant images

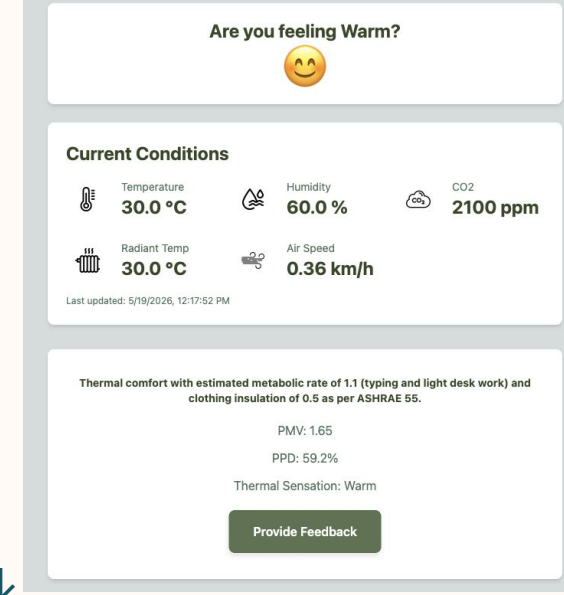


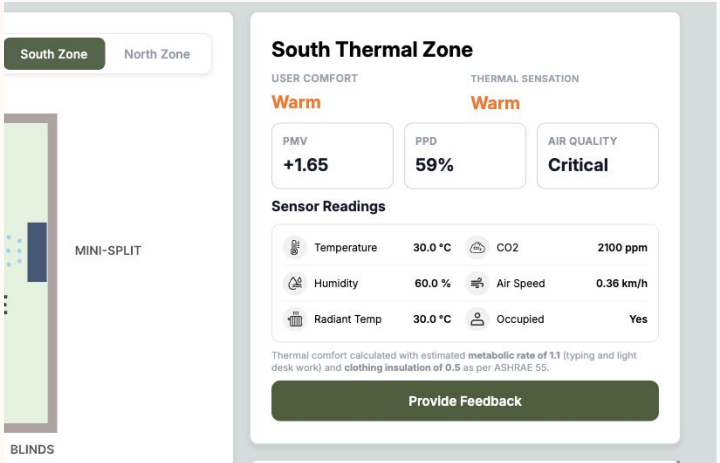
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Our Contribution

- **What was inherited**
  - Live sensor & thermal comfort data fetched from Home Assistant
  - **Before**: No data persist → No data available for client's research
  - Treated all zones the same -> No zone awareness
- **What our team improved + why better now**
  - Added new DB tables for client to research on sensor data
  - Zone-aware Revamped UI: switchable 'South/North Thermal Zone'

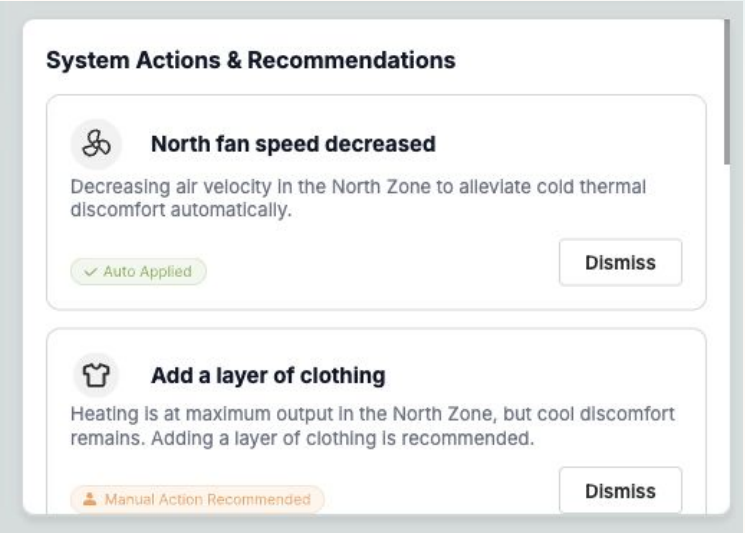




Our Contribution

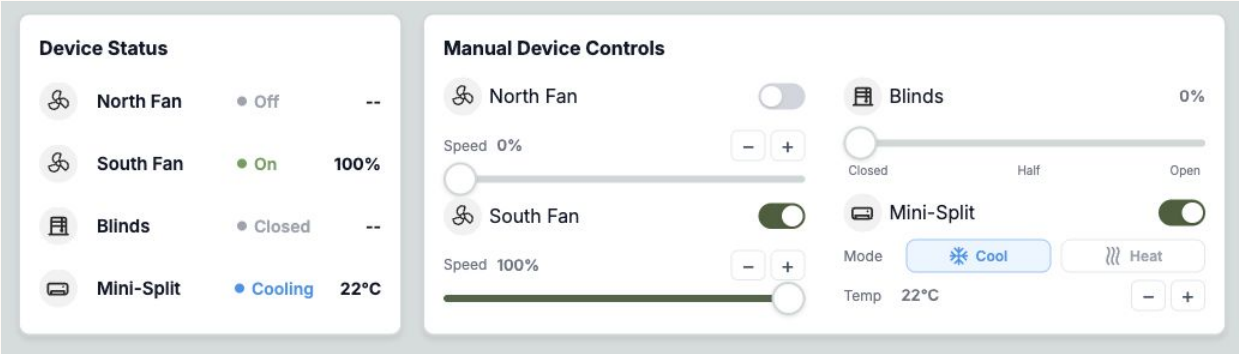
- **New features our team added**
  - System performs automated actions
  - Save & track system actions history in new DB
  - System actions & recommendations cards
- **Why better now**
  - Occupants don’t have to adjust devices
  - New DB allows client to analyze system actions history
  - Occupants see what system is doing and why

    → trust & transparency



Our Contribution

- **New features our team added**
  - Occupants manually override system actions
  - Save & track user manual override history in new DB
  - Live device (e.g. fans, blinds) status
- **Why better now**
  - Occupants verify devices are actually in expected state
  - → Trust & accuracy



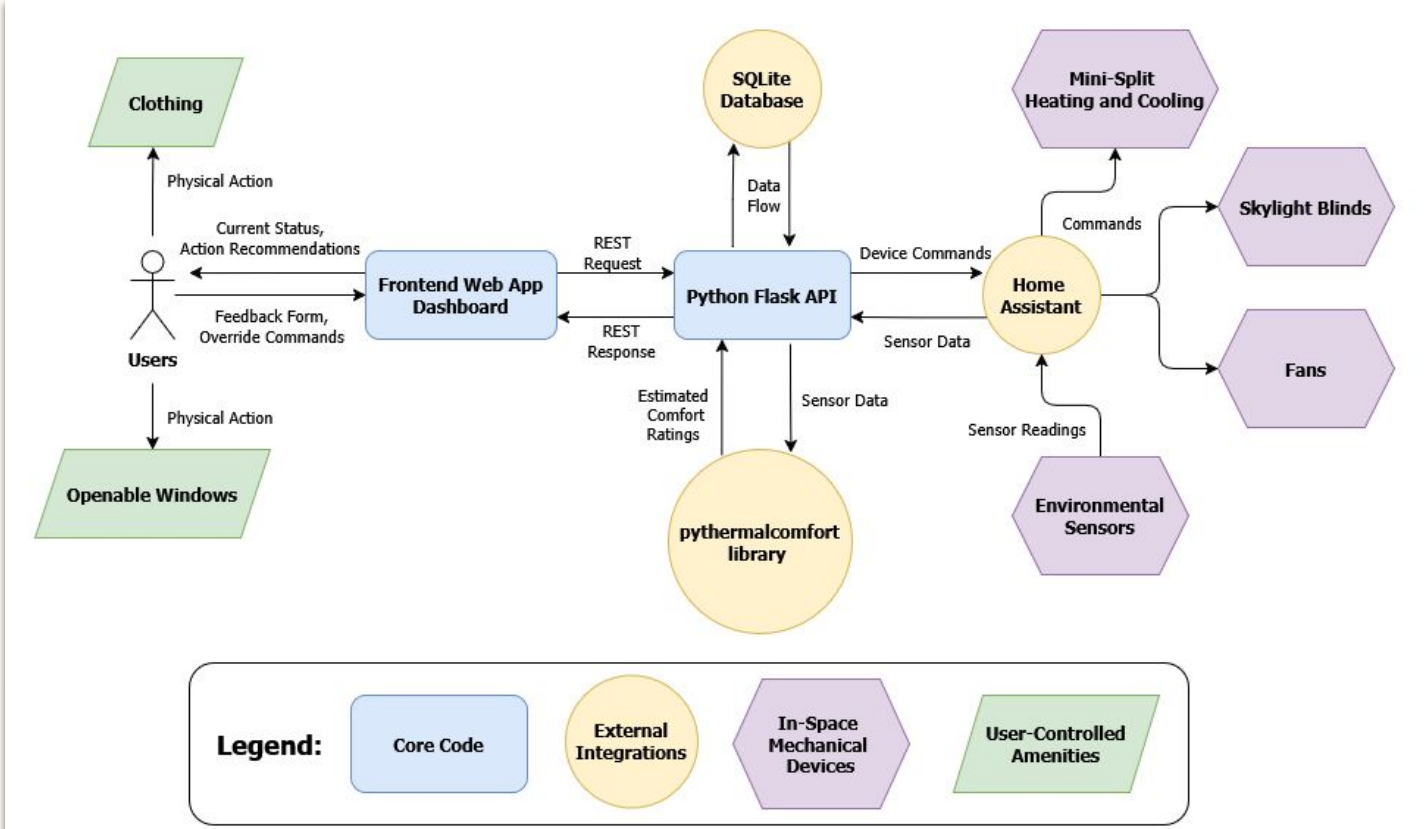
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System Architecture

System Architecture

- **Frontend** Web Application Dashboard
- **Backend** Python Flask API Server
- **Adapters**
  - Storage → Data persistence
  - CommandGenerator → Automated system actions
  - ComfortModel → Thermal comfort estimation
  - HomeAssistant → Home Assistant
- **Home Assistant** instance with mock devices

System Architecture



Technical Solution

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Technical Solution

Languages	**Backend**: Python **Frontend**: Javascript, HTML & CSS **Database**: SQL **Home Assistant Integration**: Jinja2 (templating language)
Technologies, Frameworks & Libraries	**Backend**: Flask, pydantic_settings, Click, pythermalcomfort **Frontend**: Tailwind CSS **Database**: SQLite, sqlite3 **Orchestration**: uv, Docker **IoT Integration**: Home Assistant
Support Tools	**Project Management**: Jira **Client Communication**: Slack, BCIT OneDrive **Team Communication**: Discord

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Implementation and Live Demo

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Project Management and Delivery

Project Management and Delivery

	Outcomes	Major milestones
Sprint 1	• Restructure DB schema • Revamp dashboard UI • Occupants clearly see divided zones, system actions, and manual device controls	• Created new ERD diagram with all tables have relationships each other • Created new Figma prototype • Marked up, styled & made components interactive

Project Management and Delivery

	Outcomes	Major milestones
Sprint 2	• Occupants can see live sensor & device data refreshed every minute • Occupants better experience feedback • Our app process client's new rule-based system • Occupants see personalized system actions & recommendations	Built backend endpoints to: • Fetch live sensor & device data from Home Assistant • Save feedback & Home Assistant data • Refine recommendation logic • Send back clothing layers recommendation • Send "system_actions" within response payload
Sprint 3	• Occupants manually override device • System performs automated actions • Client Handoff	• Send POST request to Home Assistant to change sensor or device status • Save sensor readings • Connect system actions & user manual override between Home Assistant and the frontend

Project Management and Delivery

Key challenges	How we addressed
Documentation Overload: over-invested time into documentation → delayed app development	• Timebox documentation • Used past sprint data to plan scopes more realistically • Prioritize improving product over writing documentation
Communication Gaps: Overlapping task ownership led to some duplicated work	• Assigned explicit ownership before starting work • Reviewed similar tasks together to prevent overlapping

Project Management and Delivery

	Planned work	Actual work	Estimated Hours	Actual hours
Sprint 1	131 (100%)	126 (96%)	200	**217.34**
Sprint 2	121 (100%)	113 (93%)	206	**239.67**
Total (Sprint 1 + 2)	252 (100%)	239 (95%)	406	**457.01**

9 Teamwork and Collaboration

Collaboration through Work Division

- Consider everyone's strengths and experiences
- Here're just some examples ...

Kai	Database option	→	Database redesign + configuration
Heather	Had frontend dev work exp	→	UI redesign
Shinyoung	Had frontend dev work exp	→	UI implementation
Khai	Used Python during co-op	→	Backend implementation (in Python)
Hei	Had admin work exp	→	Documentation formatting

Collaboration through Branching

- **Branching** strategy for source control
- Each member worked on isolated feature branches
  - **Parallel development** without interrupting each other
  - Main branch always remains safe and stable
- **Peer review** required for pull requests before merging
  - Helps catch bugs early

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Complexity, Creativity, and Innovation

Challenges

- **Undeployed system**
  - System not yet deployed to the actual office space
  - Some sensors and actuators not yet installed
  - Cannot test in an actual production environment
- **Restrictive technology stack**
  - To revamp an existing prototype developed by another CST team
  - Must deal with technologies that we are unfamiliar with

Solutions

- **Undeployed system**
  - Set up fake Home Assistant in Docker
  - Add mocked sensors and actuators to fake Home Assistant
  - Simulate an actual environment for testing
- **Restrictive technology stack**
  - Read through available documentation
  - Share insights to shorten each individual’s learning curve

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Conclusion and Next Steps

Conclusion

- **Purpose** — develop a smart environmental control system to:
  - Improve indoor thermal comfort
  - Avoid energy waste
- **Outcomes** — by the end of this project, we successfully:
  - Redesigned UI to display thermal data by zones
  - Restructured database to better support data analysis
  - Extended functionality, e.g., manual override, floorplan zone view

Next Steps

- **Production deployment**
  - Deploy prototype in production environment with actual sensors and actuators
  - Use, test, and improve concurrently while system is actually in use
- **AI integration**
  - Align with the client’s goal
  - Introduce AI to handle system automation
  - Handle how AI ties in with other system components, e.g., Home Assistant

Question and Answer

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